



Class - X

2023-24

Board: Maharashtra

LINEAR EQUATION IN TWO VARIABLES

1. LINEAR EQUATION IN TWO VARIABLES:

- An equation which contains two variables and the degree of each term containing variable is one, is called a linear equation in two variables.

2. GENERAL FORM:

- $ax + by + c = 0$ is the general form of a linear equation in two variables; a, b, c are real numbers and a, b are not equal to zero at the same time.

Example: $3x - 4y + 12 = 0$ is the general form of equation $3x = 4y - 12$.

3. SIMULTANEOUS LINEAR EQUATIONS:

- When we think about two linear equations in two variables at the same time, they are called simultaneous equations.

Example: $5x - 3y = 8$; $3x + y = 2$ are Simultaneous linear equations.

4. SOLUTION OF LINEAR EQUATION IN TWO VARIABLES:

- A solution to an equation is a value of a variable that makes a true statement when substituted into the equation.
- The process of finding the solution to an equation is called solving the equation.
- Each solution (x, y) of a linear equation in two variables, $ax + by + c = 0$, corresponds to a point on the line representing the equation, and vice versa.

Steps to find solution to an equation:

1. Substitute the number for the variable in the equation.
 2. Simplify the expressions on both sides of the equation.
 3. Determine whether the resulting equation is true.
- ◆ If it is true, the number is a solution.
 - ◆ If it is not true, the number is not a solution.

We can say 3 is a solution to the equation $x + 2 = 5$ because when we substitute 3 for x the resulting statement is true.

5. METHODS OF SOLVING LINEAR EQUATION IN TWO VARIABLES:

1. Substitution Method
2. Elimination Method
3. Determinant Method
4. Graphical Method

Elimination Method

In this method, we eliminate one of the variables to obtain equations in one variable. We can solve and find the value of one of the two variables and then substituting this value in one of the given equations we can find the value of the other variable.



Example: Solve $x + y = 14$ and $x - y = 2$.

Solution: By adding both the equations we get an equation in one variable

$$\begin{array}{rcl} x + y & = & 14 \quad \text{..... (I)} \\ + \quad x - y & = & 2 \quad \text{..... (II)} \\ \hline 2x + 0 & = & 16 \\ 2x & = & 16 \\ x & = & 8 \end{array}$$

Substituting $x = 8$ in the equation (I)

$$\begin{array}{l} x + y = 14 \\ \therefore 8 + y = 14 \\ \therefore y = 6 \end{array}$$

Here (8, 6) is the solution of first equation. Let us check, whether it satisfies the second equation also.

$$x - y = 8 - 6 = 2 \text{ is true.}$$

$\therefore$ (8, 6) is the solution for both the equations.

Hence (8, 6) is the solution of simultaneous equations $x + y = 14$ and $x - y = 2$.

Substitution Method

We can express one variable in terms of other from one of the equations. Then substituting it in the other equation we can eliminate the variable. Let us discuss this method from following examples.

Example: Solve $8x + 3y = 11$; $3x - y = 2$

$$\begin{array}{rcl} \text{Solution: } 8x + 3y & = & 11 \quad \text{.....(I)} \\ 3x - y & = & 2 \quad \text{.....(II)} \end{array}$$

In Equation (II), it is easy to express y in terms of x .

$$\begin{array}{l} 3x - y = 2 \\ 3x - 2 = y \end{array}$$

Substituting $y = 3x - 2$ in equation (I).

$$\begin{array}{l} x + 3y = 11 \\ \therefore 8x + 3(3x - 2) = 11 \\ \therefore 8x + 9x - 6 = 11 \\ \therefore 17x - 6 = 11 \\ \therefore 17x = 11 + 6 = 17 \\ \therefore x = 1 \end{array}$$

Now, substituting this value of x in the equation $y = 3x - 2$.

$$\begin{array}{l} y = 3 \times 1 - 2 \\ \therefore y = 1 \\ \therefore (1, 1) \text{ is the solution of the given equations} \end{array}$$

Determinant Method

$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ is a determinant. $(a, b), (c, d)$ are rows and $\begin{pmatrix} a \\ c \end{pmatrix}, \begin{pmatrix} b \\ d \end{pmatrix}$ are columns.

Degree of this determinant is 2, because there are 2 elements in each column and 2 elements in each row. Determinant represents a number which is $(ad - bc)$.

$$\begin{array}{c} \times \\ \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \end{array}$$

$ad - bc$ is the value of determinant $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$

Determinants, usually, are represented with capital letters as $A, B, C, D, \dots$ etc.

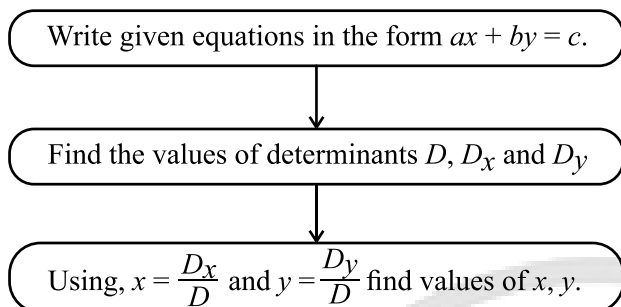
$$(1) \quad A = \begin{vmatrix} 5 & 3 \\ 7 & 9 \end{vmatrix} = (5 \times 9) - (3 \times 7) = 45 - 21 = 24$$



Determinant method (Cramer's Rule)

- Using determinants, simultaneous equations can be solved easily and in less space. This method is known as determinant method. This method was first given by a Swiss mathematician Gabriel Cramer, so it is also known as Cramer's method.

Steps:



- ◆ In D the column of constants $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$ is omitted.
- ◆ In D_x the column of the coefficients of x , $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ is replaced by $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$.
- ◆ In D_y the column of the coefficients of y , $\begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ is replaced by $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$.

Example: Solve the following simultaneous equations using Cramer's Rule.

$$5x + 3y = -11; 2x + 4y = -10$$

Solution: Given equations

$$5x + 3y = -11$$

$$2x + 4y = -10$$

$$D = \begin{vmatrix} 5 & 3 \\ 2 & 4 \end{vmatrix} = (5 \times 4) - (2 \times 3) = 20 - 6 = 14$$

$$D_x = \begin{vmatrix} -11 & 3 \\ -10 & 4 \end{vmatrix} = (-11) \times 4 - (-10) \times 3 = -44 - (-30) = -44 + 30 = -14$$

$$D_y = \begin{vmatrix} 5 & -11 \\ 2 & -10 \end{vmatrix} = 5 \times (-10) - 2 \times (-11) = -50 - (-22) = -50 + 22 = -28$$

$$x = \frac{D_x}{D} = \frac{-14}{14} = -1 \text{ and } y = \frac{D_y}{D} = \frac{-28}{14} = -2$$

∴ $(x, y) = (-1, -2)$ is the solution.

Graphical Method

In previous standard we learnt that the graph of a linear equation in two variables is a straight line.



Steps:

Step-1: To draw the graphs of the two given equations, first find three solutions of each equations and make the table for each equation.

Step-2: Now plot the points on same graph paper and join them to get the graph of each equation.

Step-3: The lines of the two given simultaneous equations will intersect each other at a point. The co-ordinates of the point of intersection of the two lines is the solution of the given simultaneous equations.

Note: Avoid two-digit coordinates and using decimal coordinates while finding ordered pairs so that plotting points becomes easy.

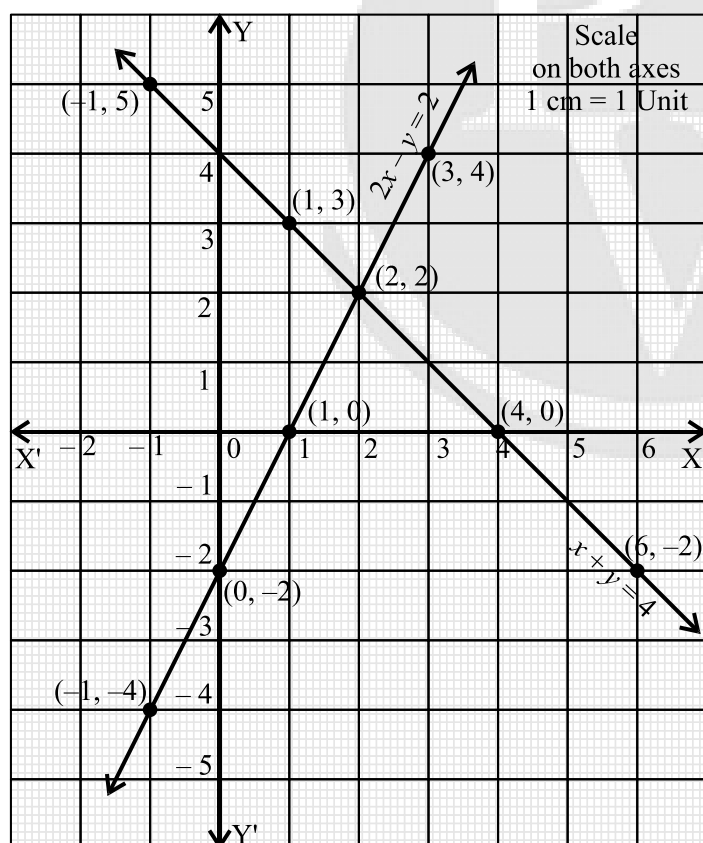
Example: Let's draw graphs of $x + y = 4$, $2x - y = 2$ and observe them.

$$x + y = 4$$

x	-1	4	1	6
y	5	0	3	-2
(x, y)	(-1, 5)	(4, 0)	(1, 3)	(6, -2)

$$2x - y = 2$$

x	0	1	3	-1
y	-2	0	4	-4
(x, y)	(0, -2)	(1, 0)	(3, 4)	(-1, -4)



Each point on the graph satisfies the equation. The two lines intersect each other at (2, 2).

Hence ordered pair (2, 2) i.e. $x = 2$, $y = 2$ satisfies the equations $x + y = 4$ and $2x - y = 2$.

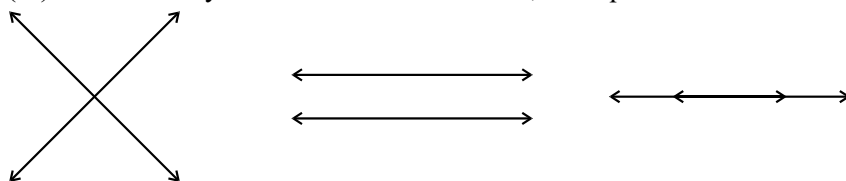
The values of variables that satisfy the given equations, give the solution of given equations.

∴ The solution of given equations $x + y = 4$, $2x - y = 2$ is $x = 2$, $y = 2$.

- Since (2, 2) is the only common point on both the lines, there is one and only one solution for this pair of linear equations in two variables but sometimes line may be parallel or coincident.
- So we can summarise the behaviour of lines representing a pair of linear equations in two variables as follows:



- (i) the lines may intersect in a single point. In this case, the pair of equations has a unique solution.
- (ii) the lines may be parallel. In this case, the equations have no solution.
- (iii) the lines may be coincident. In this case, the equations have infinitely many solutions.



➤ Lines represented by the equations.

$$a_1x + b_1y + c_1 = 0, \quad a_2x + b_2y + c_2 = 0$$

are (i) intersecting, then $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$.

(ii) coincident, then $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

(iii) parallel, then $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

6. EQUATIONS REDUCIBLE TO LINEAR EQUATIONS IN TWO VARIABLES:

We can create new variables making a proper change in the given variables. Substituting the new variables in the given non-linear equations, we can convert them in linear equations.

Example: $\frac{4}{x} + \frac{5}{y} = 7$; $\frac{3}{x} + \frac{4}{y} = 5$.

Solution: $\frac{4}{x} + \frac{5}{y} = 7$; $\frac{3}{x} + \frac{4}{y} = 5$.

$$4\left(\frac{1}{x}\right) + 5\left(\frac{1}{y}\right) = 7 \quad \text{..... (I)}$$

$$3\left(\frac{1}{x}\right) + 4\left(\frac{1}{y}\right) = 5 \quad \text{..... (II)}$$

Replacing $\left(\frac{1}{x}\right)$ by m and $\left(\frac{1}{y}\right)$ by n in equations (I) and (II), we get

$$4m + 5n = 7 \quad \text{..... (III)}$$

$$3m + 4n = 5 \quad \text{..... (IV)}$$

On solving these equations we get

$$m = 3, n = -1$$

$$\text{Now, } m = \frac{1}{x} \quad \therefore 3 = \frac{1}{x} \quad \therefore x = \frac{1}{3}$$

$$n = \frac{1}{y} \quad \therefore -1 = \frac{1}{y} \quad \therefore y = -1$$

$\therefore$ Solution of given simultaneous equations is $(x, y) = \left(\frac{1}{3}, -1\right)$

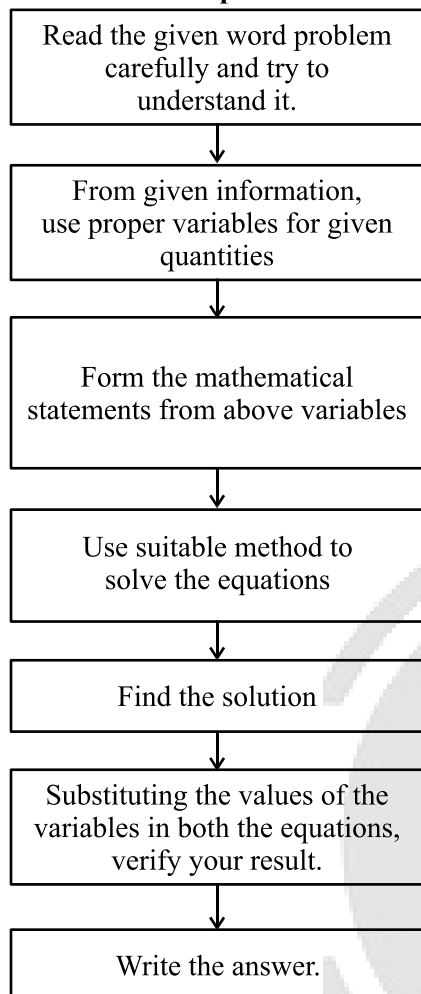
7. APPLICATION OF SIMULTANEOUS EQUATION:

Word problems based on simultaneous equations:

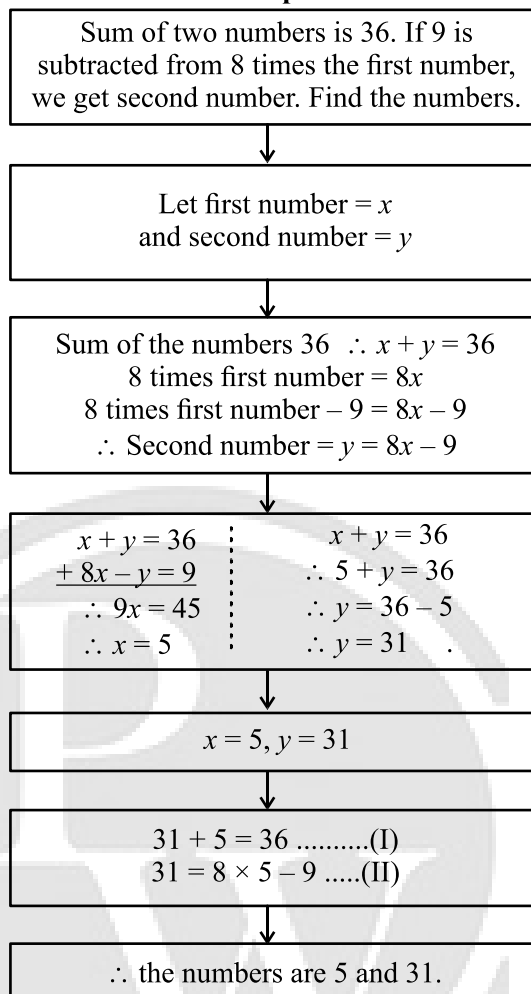
While solving word problems, converting the given information into mathematical form is an important step in this process. In the following flow-chart, the procedure for finding solutions of word problems is given.



Steps



Example





Class - X

2023-24

Board: Maharashtra

QUADRATIC EQUATION

1. STANDARD FORM OF QUADRATIC EQUATION:

- The equation involving one variable and having 2 as the maximum index of the variable is called the quadratic equation.

General form is $ax^2 + bx + c = 0$

In $ax^2 + bx + c = 0$, a, b, c are real numbers and $a \neq 0$.

Example:

Decide which of the following are quadratic equations?

- (1) $3x^2 - 5x + 3 = 0$
- (2) $9y^2 + 5 = 0$
- (3) $m^3 - 5m^2 + 4 = 0$
- (4) $(l + 2)(l - 5) = 0$

Solution:

- (1) In the equation $3x^2 - 5x + 3 = 0$, x is the only variable and maximum index of the variable is 2
 $\therefore$ It is a quadratic equation.
- (2) In the equation $9y^2 + 5 = 0$, y is the only variable and maximum index of the variable is 2
 $\therefore$ It is a quadratic equation.
- (3) In the equation $m^3 - 5m^2 + 4 = 0$, m is the only variable but maximum index of the variable is not 2.
 $\therefore$ It is not a quadratic equation.
- (4) $(l + 2)(l - 5) = 0$
 $\therefore l(l - 5) + 2(l - 5) = 0$
 $\therefore l^2 - 5l + 2l - 10 = 0$
 $\therefore l^2 - 3l - 10 = 0$, In this equation l is the only variable and maximum index of the variable is 2.
 $\therefore$ It is a quadratic equation.

2. ROOTS OF QUADRATIC EQUATION:

- The values of variable which satisfy the equation [or the value for which both the sides of equation are equal] are called solutions or roots of the equation.

Example:

$2x^2 - 7x + 6 = 0$ check whether (i) $x = \frac{3}{2}$, (ii) $x = -2$ are solutions of the equations.

Solution:

- (i) Put $x = \frac{3}{2}$ in the polynomial $2x^2 - 7x + 6$

$$2x^2 - 7x + 6 = 2\left(\frac{3}{2}\right)^2 - 7\left(\frac{3}{2}\right) + 6$$

$$= 2 \times \frac{9}{4} - \frac{21}{2} + 6$$

$$= \frac{9}{2} - \frac{21}{2} + \frac{12}{2} = 0$$

$$\therefore x = \frac{3}{2} \text{ is a solution of the equation.}$$



- (ii) Let $x = -2$ in $2x^2 - 7x + 6$
 $2x^2 - 7x + 6 = 2(-2)^2 - 7(-2) + 6$
 $= 2 \times 4 + 14 + 6$
 $= 28 \neq 0$
 $\therefore x = -2$ is not a solution of the equation.

Activity:

If $x = 5$ is a root of equation $kx^2 - 14x - 5 = 0$ then find the value of k by completing the following activity.

Solution:

One of the roots of equation $kx^2 - 14x - 5 = 0$ is 5.

$\therefore$ Now Let $x = 5$ in the equation.

$$k[5]^2 - 14[5] - 5 = 0$$

$$\therefore 25k - 70 - 5 = 0$$

$$25k - [75] = 0$$

$$25k = [75]$$

$$\therefore k = \frac{[75]}{[25]} = 3$$

3. SOLUTIONS OF A QUADRATIC EQUATION BY FACTORISATION:

- By substituting arbitrary values for the variable and deciding the roots of quadratic equation is a time consuming process. Let us learn to use factorisation method to find the roots of the given quadratic equation.

$$x^2 - 4x - 5 = (x - 5)(x + 1)$$

$(x - 5)$ and $(x + 1)$ are two linear factors of quadratic polynomial $x^2 - 4x - 5$.

If product of two numbers is zero then at least one of them is zero.

$$\therefore x - 5 = 0 \text{ or } x + 1 = 0$$

$$\therefore x = 5 \text{ or } x = -1$$

$\therefore 5$ and the -1 are the roots of the given quadratic equation.

While solving the equation first we obtained the linear factors. So we call this method as 'factorization method' of solving quadratic equation.

Example:

Solve the following quadratic equations by factorisation.

(1) $m^2 - 14m + 13 = 0$

$$\therefore m^2 - 13m - 1m + 13 = 0$$

$$\therefore \overline{-m(m-13)} - \overline{1(m-13)} = 0$$

$$\therefore (m-13)(m-1) = 0$$

$$\therefore m - 13 = 0 \text{ or } m - 1 = 0$$

$$\therefore m = 13 \text{ or } m = 1$$

$\therefore 13$ and 1 are the roots of the given quadratic equation.

(2) $3y^2 = 15y$

$$\therefore 3y^2 - 15y = 0$$

$$\therefore 3y(y-5) = 0$$

$$\therefore 3y = 0 \text{ or } (y-5) = 0$$

$$\therefore y = 0 \text{ or } y = 5$$

$\therefore 0$ and 5 are the roots of quadratic equation.



$$\begin{aligned}(3) \quad & x^2 = 3 \\ & \therefore x^2 - 3 = 0 \\ & \therefore x^2 - (\sqrt{3})^2 = 0 \\ & \therefore (x + \sqrt{3})(x - \sqrt{3}) = 0 \\ & \therefore (x + \sqrt{3}) = 0 \text{ or } (x - \sqrt{3}) = 0 \\ & \therefore x = -\sqrt{3} \text{ or } x = \sqrt{3} \\ & \therefore -\sqrt{3} \text{ and } \sqrt{3} \text{ are the roots of given quadratic equation.}\end{aligned}$$

$$\begin{aligned}(4) \quad & 6\sqrt{3}x^2 + 7x = \sqrt{3} \\ & \therefore 6\sqrt{3}x^2 + 7x - \sqrt{3} = 0 \\ & \therefore 6\sqrt{3}x^2 + 9x - 2x - \sqrt{3} = 0 \\ & \therefore 3\sqrt{3}x(2x + \sqrt{3}) - 1(2x + \sqrt{3}) = 0 \\ & \therefore (2x + \sqrt{3})(3\sqrt{3}x - 1) = 0 \\ & \therefore 2x + \sqrt{3} = 0 \text{ or } 3\sqrt{3}x - 1 = 0 \\ & \therefore 2x = -\sqrt{3} \text{ or } 3\sqrt{3}x = 1 \\ & \therefore x = -\frac{\sqrt{3}}{2} \text{ or } x = \frac{1}{3\sqrt{3}} \\ & 6\sqrt{3} \times -\sqrt{3} = -18 \\ & \quad \quad \quad -18 \\ & \quad \quad \quad \wedge \\ & \quad \quad \quad 9 \quad -2 \\ & 9 = 3\sqrt{3} \times \sqrt{3} \\ & \therefore -\frac{\sqrt{3}}{2} \text{ and } \frac{1}{3\sqrt{3}} \text{ are the roots of the given quadratic equation.}\end{aligned}$$

4. SOLUTION OF A QUADRATIC EQUATION BY COMPLETING THE SQUARE:

Stape-1 Convert the coefficient of x^2 into 1.

Stape-2 Shift constant term to the right side.

Stape-3 Find third term using formula $= \left(\frac{1}{2} \text{coef. } x \right)^2$.

Stape-4 Add third term on both sides.

Stape-5 Complete the formula of square.

Stape-6 Take square root on both sides.

Stape-7 Find the value of x .



Example:

Solve the following quadratic equations by completing the square method.

$$9y^2 - 12y + 2 = 0$$

Solution:

Divide given equation by 9

$$y^2 - \frac{4}{3}y + \frac{2}{9} = 0 \qquad y^2 - \frac{4}{3}y = -\frac{2}{9}$$

$$\text{Third term} = \left(\frac{1}{2} \text{coef. } y\right)^2$$

$$= \left(\frac{1}{2} \times -\frac{4}{3}\right)^2 = \left(\frac{-2}{3}\right)^2 = \frac{4}{9}$$

$$y^2 - \frac{4}{3}y + \frac{4}{9} = -\frac{2}{9} + \frac{4}{9}$$

$$\left(y - \frac{2}{3}\right)^2 = \frac{2}{9}$$

$$y - \frac{2}{3} = \pm \frac{\sqrt{2}}{3} \dots\dots\dots \text{(Taking square root)}$$

$$y - \frac{2}{3} = \frac{\sqrt{2}}{3} \text{ or } y - \frac{2}{3} = -\frac{\sqrt{2}}{3}$$

$$y = \frac{\sqrt{2}}{3} + \frac{2}{3} \text{ or } y = -\frac{\sqrt{2}}{3} + \frac{2}{3}$$

$$\boxed{y = \frac{2 + \sqrt{2}}{3}} \text{ or } \boxed{y = \frac{2 - \sqrt{2}}{3}}$$

5. FORMULA FOR SOLVING A QUADRATIC EQUATION:

➤ $ax^2 + bx + c = 0 \dots\dots\dots \text{(I)}$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \dots\dots\dots \text{Dividing both sides by } a.$$

$$\therefore x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 + \frac{c}{a} = 0$$

$$\therefore \left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2} + \frac{c}{a} = 0$$

$$\therefore \left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a^2} = 0 \qquad \therefore \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

$$\therefore x + \frac{b}{2a} = \sqrt{\frac{b^2 - 4ac}{4a^2}} \text{ or } x + \frac{b}{2a} = -\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$\therefore x = -\frac{b}{2a} + \sqrt{\frac{b^2 - 4ac}{4a^2}} \text{ or } x = -\frac{b}{2a} - \sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$\therefore x = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ or } x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$



In short the solution is written as $x = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$ and these values are denoted by α, β .

$$\therefore \alpha = \frac{b + \sqrt{b^2 - 4ac}}{2a}, \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a} \dots\dots\dots (I)$$

➤ Solve quadratic equations using formula.

Example:

$$m^2 - 14m + 13 = 0$$

Solution:

$$m^2 - 14m + 13 = 0 \text{ comparing with } ax^2 + bx + c = 0$$

we get $a = 1, b = -14, c = 13$,

$$\therefore b^2 - 4ac = (-14)^2 - 4 \times 1 \times 13$$

$$= 196 - 52 = 144$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{(14) \pm \sqrt{144}}{2 \times 1} = \frac{14 \pm 12}{2}$$

$$\therefore m = \frac{14 + 12}{2} \text{ or } m = \frac{14 - 12}{2}$$

$$\therefore m = \frac{26}{2} \text{ or } m = \frac{2}{2}$$

$$\therefore m = 13 \text{ or } m = 1$$

$\therefore 13$ and 1 are roots of the equation.

Example:

$$x^2 + 10x + 2 = 0$$

Solution:

$$x^2 + 10x + 2 = 0 \text{ comparing with } ax^2 + bx + c = 0$$

we get $a = 1, b = 10, c = 2$,

$$\therefore b^2 - 4ac = (10)^2 - 4 \times 1 \times 2$$

$$= 100 - 8$$

$$= 92$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-10 \pm \sqrt{92}}{2 \times 1}$$

$$x = \frac{-10 \pm \sqrt{4 \times 23}}{2}$$

$$= \frac{-10 \pm 2\sqrt{23}}{2} = \frac{-2(5 \pm \sqrt{23})}{2}$$

$$\therefore x = -(5 \pm \sqrt{23})$$

$$\therefore x = -5 + \sqrt{23} \text{ or } x = -5 - \sqrt{23}$$

$\therefore$ the roots of the given quadratic equation are $-5 + \sqrt{23}$ and $-5 - \sqrt{23}$.



6. NATURE OF ROOTS OF A QUADRATIC EQUATION:

➤ You know that $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ are roots of quadratic equation $ax^2 + bx + c = 0$

(1) If $b^2 - 4ac = 0$ then, $x = \frac{-b \pm \sqrt{0}}{2a} \therefore x = \frac{-b+0}{2a}$ or $x = \frac{-b-0}{2a}$

$\therefore$ the roots of the quadratic equation are real and equal.

(2) If $b^2 - 4ac > 0$ then, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

i.e. $x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$

$\therefore$ roots of the quadratic equation are real and unequal.

(3) If $b^2 - 4ac < 0$ then, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ are not real numbers

$\therefore$ the roots of quadratic equations are not real.

Nature of roots of quadratic equation is determined by the value of $b^2 - 4ac$. $b^2 - 4ac$ is called discriminant of a quadratic equation and is denoted by Greek letter Δ (Delta)

Example:

Find the value of the discriminant of the equation $x^2 + 10x - 7 = 0$

Solution:

Comparing $x^2 + 10x - 7 = 0$ with $ax^2 + bx + c = 0$.

$a=1, b=10, c=-7,$

$\therefore b^2 - 4ac = 10^2 - 4 \times 1 \times -7$

$= 100 + 28$

$= 128$

Example:

Determine nature of roots of the quadratic equations.

(i) $2x^2 - 5x + 7 = 0$

Solution:

Compare $2x^2 - 5x + 7 = 0$ with $ax^2 + bx + c = 0$.

$ax^2 + bx + c = 0$

$a=2, b=-5, c=7,$

$\therefore b^2 - 4ac = (-5)^2 - 4 \times 2 \times 7$

$D = 25 - 56 \quad D = -31$

$\therefore b^2 - 4ac < 0$

$\therefore$ the roots of the equation are not real.

Example:

Determine nature of roots of the quadratic equations.

(ii) $2x^2 + 2x - 7 = 0$

Solution:

Compare $x^2 + 2x - 9 = 0$ with $ax^2 + bx + c = 0$.

$a = \boxed{1}, b = 2, c = \boxed{-9},$

$\therefore b^2 - 4ac = 2^2 - 4 \times \boxed{1} \times \boxed{-9}$

$D = 4 - \boxed{36} \quad D = 40$

$\therefore b^2 - 4ac > 0$

$\therefore$ the roots of the equation are real and unequal.



7. THE RELATION BETWEEN ROOTS OF THE QUADRATIC EQUATION AND COEFFICIENTS:

➤ α and β are the roots of the equation $ax^2 + bx + c = 0$,

$$\begin{aligned}\alpha + \beta &= \frac{-b + \sqrt{b^2 - 4ac}}{2a} + \frac{-b - \sqrt{b^2 - 4ac}}{2a} \\&= \frac{-b + \sqrt{b^2 - 4ac} - b - \sqrt{b^2 - 4ac}}{2a} \\&= -\frac{2b}{2a}\end{aligned}$$

$$\therefore \boxed{\alpha + \beta = -\frac{b}{a}}$$

$$\begin{aligned}\alpha \times \beta &= \frac{-b + \sqrt{b^2 - 4ac}}{2a} \times \frac{-b - \sqrt{b^2 - 4ac}}{2a} \\&= \frac{(-b + \sqrt{b^2 - 4ac})(-b - \sqrt{b^2 - 4ac})}{4a^2} \\&= \frac{b^2 - (b^2 - 4ac)}{4a^2} \\&= \frac{4ac}{4a^2} = \frac{c}{a}\end{aligned}$$

$$\therefore \boxed{\alpha \beta = \frac{c}{a}}$$

Example:

If α and β are the roots of the quadratic equation $2x^2 + 6x - 5 = 0$, then find $(\alpha + \beta)$ and $\alpha \times \beta$.

Solution:

Comparing $2x^2 + 6x - 5 = 0$ with $ax^2 + bx + c = 0$.

$$\therefore a = 2, b = 6, c = -5$$

$$\therefore \alpha + \beta = -\frac{b}{a} = -\frac{6}{2} = -3$$

$$\text{and } \alpha \times \beta = \frac{c}{a} = \frac{-5}{2}$$

Example:

If α and β are the roots of $x^2 + 5x - 1 = 0$ then find –

$$(i) \quad \alpha^3 + \beta^3 \quad (ii) \quad \alpha^2 + \beta^2.$$

Solution:

$$x^2 + 5x - 1 = 0$$

$$a = 1, b = 5, c = -1$$

$$\alpha + \beta = -\frac{b}{a} = \frac{-5}{1} = -5$$

$$\alpha \times \beta = \frac{c}{a} = \frac{-1}{1} = -1$$



$$(i) \quad \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

$$= (-5)^3 - 3 \times (-1) \times (-5)$$

$$= -125 - 15$$

$$\alpha^3 + \beta^3 = -140$$

$$(ii) \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= (-5)^2 - 2 \times (-1)$$

$$= 25 + 2$$

$$\alpha^2 + \beta^2 = 27$$

8. TO OBTAIN A QUADRATIC EQUATION HAVING GIVEN ROOTS:

- Let α and β be the roots of a quadratic equation in variable x .

$$\therefore x = \alpha \text{ or } x = \beta$$

$$\therefore x - \alpha = 0 \text{ or } x - \beta = 0$$

$$\therefore (x - \alpha)(x - \beta) = 0$$

$$\therefore x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

When two roots of equation are given then quadratic equation can be obtained as $x^2 - (\text{addition of roots})x + \text{product of the roots} = 0$.

Example:

Obtain the quadratic equation if roots are $-3, -7$.

Solution:

$$\text{Let } \alpha = -3 \text{ and } \beta = -7$$

$$\therefore \alpha + \beta = (-3) + (-7) = -10 \text{ and } \alpha \times \beta = (-3) \times (-7) = 21$$

$$\therefore \text{and quadratic equation is, } x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\therefore x^2 - (-10)x + 21 = 0$$

$$\therefore x^2 + 10x + 21 = 0$$

9. APPLICATION OF QUADRATIC EQUATION:

- Quadratic equations are useful in daily life for finding solutions of some practical problems. We are now going to learn the same. With the help of an example.

Example:

A train travels 360 km with uniform speed. The speed of the train is increased by 5 km/hr, it takes 48 minutes less to cover the same distance. Find the initial speed of the train.

Solution:

Let initial speed of the train be x km/hr.

$\therefore$ New speed is $(x + 5)$ km/hr.

$$\text{time to cover 360 km} = \frac{\text{distance}}{\text{speed}} = \frac{360}{x} \text{ hours.}$$

$$\text{New time after increasing speed} = \frac{360}{x + 5} \text{ hours. from given condition}$$



$$\begin{aligned}\frac{360}{x+5} &= \frac{360}{x} - \frac{48}{60} \quad \dots\dots\dots \left(48 \text{ min} = \frac{48}{60} \text{ hrs} \right) \\ \therefore \frac{360}{x} - \frac{360}{x+5} &= \frac{48}{60} \\ \therefore \frac{1}{x} - \frac{1}{x+5} &= \frac{48}{60 \times 360} \quad \dots\dots\dots (\text{Dividing both sides by } 360) \\ \therefore \frac{x+5x}{x(x+5)} - \frac{4}{5 \times 360} & \\ \therefore \frac{5}{x^2+5x} - \frac{1}{5 \times 90} & \\ \therefore \frac{5}{x^2+5x} + \frac{1}{450} & \\ \therefore x^2+5x &= 2250 \\ \therefore x^2+5x-2250 &= 0 \\ \therefore x^2+50x-45x-2250 &= 0 \\ \therefore x(x+50)-45(x+50) &= 0 \\ \therefore (x+50)(x-45) &= 0 \\ &\quad -2250 \\ &\quad / \quad \backslash \\ +50 &\quad -45 \\ \therefore x+50 &= 0 \text{ or } x-45 = 0 \\ \therefore x &= -50 \text{ or } x = 45 \\ \text{But speed is never negative } \therefore x &\neq -50 \\ \therefore x &= 45 \\ \therefore \text{Initial speed of the train is } &45 \text{ km/hr.}\end{aligned}$$





2023-24

Class - X

Board: Maharashtra

ARITHMETIC PROGRESSION

1. SEQUENCE:

- A set of numbers where the numbers are arranged in a definite order, like the natural numbers, is called a sequence.

Terms in a sequence

- In a sequence, ordered terms are represented as $t_1, t_2, t_3, \dots, t_n \dots$. In general sequence is written as $\{t_n\}$. If the sequence is infinite, for every positive integer n , there is a term t_n .

2. ARITHMETIC PROGRESSION:

- In the given sequence if the difference between two consecutive terms ($t_{n+1} - t_n$) is constant then the sequence is called Arithmetic Progression (A.P.). In this sequence $t_{n+1} - t_n = d$ is the common difference.

Difference d can be positive, negative or zero.

In an A.P. if first term is denoted by a and common difference is d then, $t_1 = a, t_2 = a + d$.

$$t_3 = (a + d) + d = a + 2d$$

A.P. having first term as a and common difference d is $a, (a+d), (a+2d), (a+3d), \dots$

A.P. can be finite or infinite

Example:

Which of the following sequences are A.P? If it is an A.P, find next two terms.

- (i) 5, 12, 19, 26, ...
- (ii) 2, -2, -6, -10, ...

Solution: (i)

In this sequence 5, 12, 19, 26, ...,

First term $= t_1 = 5, \quad t_2 = 12, \quad t_3 = 19, \dots$

$$t_2 - t_1 = 12 - 5 = 7$$

$$t_3 - t_2 = 19 - 12 = 7$$

Here first term is 5 and common difference which is constant is $d = 7$

$\therefore$ This sequence is an A.P.

Next two terms in this A.P. are $26 + 7 = 33$ and $33 + 7 = 40$.

Next two terms in given A.P. are 33 and 40,

Solution: (ii)

In the sequence 2, -2, -6, -10,

$$t_1 = 2, \quad t_2 = -2, \quad t_3 = -6, \quad t_4 = -10 \dots\dots$$

$$t_2 - t_1 = -2 - 2 = -4$$

$$t_3 - t_2 = -6 - (-2) = -6 + 2 = -4$$

$$t_4 - t_3 = -10 - (-6) = -10 + 6 = -4$$

From this difference between two consecutive terms that is $t_n - t_{n-1} = -4$

$\therefore d = -4$, which is constant $\therefore$ It is an A.P.

Next two terms in this A.P. are $(-10) + (-4) = -14$ and $(-14) + (-4) = -18$



3. n^{th} TERM OF AN A.P.:

- In the sequence 5, 8, 11, 14, ... the difference between two consecutive terms is 3. Hence, this sequence is an A.P.

Here the first term is 5. If 3 is added to 5 we get the second term 8. Similarly to find 100th term what should be done?

First term Second term Third term ...

Number 5, $5 + 3 = 8$ $8 + 3 = 11 \dots$

In this way reaching up to 100th term will be time consuming. Let's see if we can find any formula for it.

5	8	11	14			
5	$5 + 1 \times 3$	$5 + 2 \times 3$	$5 + 3 \times 3$	$5 + (n - 1) \times 3$	$5 + n \times 3$	...
1 st term t_1	2 nd term t_2	3 rd term t_3	4 th term t_4	n^{th} term t_n	$(n + 1)^{\text{th}}$ term t_{n+1}	...

Generally in the A.P. $t_1, t_2, t_3, \dots$ If first term is a and common difference is d ,

$$t_1 = a$$

$$t_2 = t_1 + d = a + d = a + (2 - 1)d$$

$$t_3 = t_2 + d = a + d + d = a + 2d = a + (3 - 1)d$$

$$t_4 = t_3 + d = a + 2d + d = a + 3d = a + (4 - 1)d$$

We get $t_n = a + (n - 1)d$.

Using the above formula we can find the 100th term of the A.P. 5, 8, 11, 14, ...

Here $a = 5, d = 3$

$$t_n = a + (n - 1)d$$

$$\therefore t_{100} = 5 + (100 - 1) \times 3$$

$$= 5 + 99 \times 3$$

$$= 5 + 297$$

$$t_{100} = 302$$

100th term of this A.P. is 302.

Let's check whether 61 is in this A.P. To find the answer we use the same formula.

$$t_n = a + (n - 1)d$$

$$t_n = 5 + (n - 1) \times 3$$

If 61 is n^{th} term means t_n , then

$$\therefore 61 = 5 + 3n - 3$$

$$= 3n + 2$$

$$\therefore 3n = 59$$

$$\therefore n = \frac{59}{3}$$

But then, n is not a natural number.

$\therefore$ 61 is not in this A.P.

Example:

Find t_n for following A.P. and then find 30th term of A.P.

3, 8, 13, 18,

Solution:

Given A.P. 3, 8, 13, 18,

Here $t_1 = 3, t_2 = 8, t_3 = 13, t_4 = 18, \dots$

$$d = t_2 - t_1 = 8 - 3 = 5$$



We know that $t_n = a + (n - 1)d$
 $\therefore t_n = 3 + (n - 1) \times 5 \because a = 3, d = 5$
 $\therefore t_n = 3 + 5n - 5$
 $\therefore t_n = 5n - 2$
 $\therefore 30^{\text{th}} \text{ term} = t_{30} = 5 \times 30 - 2$
 $= 150 - 2 = 148$

Example:

Which term of the following A.P. is 560?

2, 11, 20, 29,

Solution:

Given A.P. 2, 11, 20, 29,

Here $a = 2, d = 11 - 2 = 9$

n^{th} term of this A.P. is 560.

$$t_n = a + (n - 1)d$$

$$\therefore 560 = 2 + (n - 1) \times 9$$

$$= 2 + 9n - 9$$

$$\therefore 9n = 567$$

$$\therefore n = \frac{567}{9} = 63$$

$\therefore 63^{\text{rd}}$ term of given A.P. is 560.

4. SUM OF FIRST n TERMS OF AN A.P.:

➤ Arithmetic Progression $a, a + d, a + 2d, a + 3d, \dots, a + (n - 1)d$.

In this progression a is the first term and d is the common difference. Let's write the sum of first n terms as S_n

$$S_n = [a] + [a + d] + \dots + [a + (n - 2)d] + [a + (n - 1)d]$$

Reversing the terms and rewriting the expression again,

$$S_n = [a + (n - 1)d] + [a + (n - 2)d] + \dots + [a + d] + [a]$$

On adding,

$$2S_n = [a + a + (n - 1)d] + [a + d + a + (n - 2)d] + \dots + [a + (n - 2)d + a + d] + [a + (n - 1)d + a]$$

$$2S_n = [2a + (n - 1)d] + [2a + (n - 1)d] + \dots + [2a + (n - 1)d] \dots n \text{ times}$$

$$\therefore 2S_n = n[2a + (n - 1)d]$$

$$\therefore S_n = \frac{n}{2}[2a + (n - 1)d] \text{ or } S_n = na + \frac{n(n - 1)}{2}d$$

Example:

Let's find the sum of first 100 terms of A.P. 14, 16, 18,

Here $a = 14, d = 2, n = 100$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$\therefore S_{100} = \frac{100}{2}[2 \times 14 + (100 - 1) \times 2]$$

$$= 50[28 + 198]$$

$$= 50 \times 226 = 11300$$

$\therefore$ Sum of first 100 terms of given A.P. is 11,300.

Let's find one more formula for sum of first n terms.



In the A.P. $a, a + d, a + 2d, a + 3d, \dots, a + (n - 1)d$.

First term $= t_1 = a$ and n^{th} term is $[a + (n - 1)d]$

$$\text{Now } S_n = \frac{n}{2} [a + a + (n - 1)d]$$

$$\therefore S_n = \frac{n}{2} [t_1 + t_n] = \frac{n}{2} [\text{First term} + \text{last term}]$$

Example:

Find the sum of first n natural numbers.

Solution:

First n natural numbers are $1, 2, 3, \dots, n$.

Here $a = 1, d = 1, n^{\text{th}}$ term $= n$

$$\therefore S_n = 1 + 2 + 3 + \dots + n$$

$$S_n = \frac{n}{2} [\text{First term} + \text{last term}] \dots \text{(by the formula)}$$

$$= \frac{n}{2} [1 + n] = \frac{n(n+1)}{2}$$

$$\therefore \text{Sum of first } n \text{ natural number is } \frac{n(n+1)}{2}.$$

Example:

Find the sum of first n even natural numbers.

Solution:

First n even natural numbers are $2, 4, 6, 8, \dots, 2n$.

$t_1 = \text{First term} = 2, t_n = \text{last term} = 2n$.

Method I

$$= \frac{n}{2} [t_1 + t_n]$$

$$= \frac{n}{2} [2 + 2n]$$

$$= \frac{n}{2} \times 2(1 + n)$$

$$= n(1 + n)$$

Method II

$$S_n = 2 + 4 + 6 + 2n$$

$$= 2(1 + 2 + 3 + \dots + n)$$

$$= \frac{2[n(n+1)]}{2}$$

$$= n(1 + n)$$

Method III

$$S_n = \frac{n}{2} [2a + (n - 1)d]$$

$$= \frac{n}{2} [2 \times 2 + (n - 1)2]$$

$$= \frac{n}{2} [4 + 2n - 2]$$

$$= \frac{n}{2} [2 + 2n]$$

$$= \frac{n}{2} \times 2(1 + n)$$

$$= n(1 + n)$$

$$\therefore \text{Sum of first } n \text{ even natural numbers is } n(1 + n).$$

5. APPLICATION OF A.P.:

Example:

A mixer manufacturing company manufactured 600 mixers in 3rd year and in 7th year they manufactured 700 mixers. If every year there is same growth in the production of mixers then find.

- Production in the first year
- Production in 10th year.
- Total production in first seven years.



Solution:

Addition in the number of mixers manufactured by the company per year is constant therefore the number of production in successive years is in A.P.

(i) Let's assume that company manufactured t_n mixers in the n^{th} year then as per given information,

$$t_3 = 600, t_7 = 700$$

$$\text{We know that } t_n = a + (n - 1)d$$

$$t_3 = a + (3 - 1)d$$

$$a + 2d = 600 \dots\dots (I)$$

$$t_7 = a + (7 - 1)d$$

$$t_7 = a + 6d = 700$$

$$a + 2d = 600 \therefore \text{Substituting } a = 600 - 2d \text{ in equation (II)}$$

$$600 - 2d + 6d = 700$$

$$4d = 100 \therefore d = 25$$

$$a + 2d = 600 \therefore a + 2 \times 25 = 600$$

$$a + 50 = 600 \therefore a = 550$$

$\therefore$ Production in first year was 550.

$$(ii) t_n = a + (n - 1)d$$

$$t_{10} = 550 + (10 - 1) \times 25$$

$$= 550 + 225$$

Production in 10th year was 775.

Example:

In year 2015, Mrs. Shaikh got a job with salary ₹1,80,000 per year. Her employer agreed to give ₹10,000 per year as increment. Then in how many years will her annual salary be ₹2,50,000?

Solution:

Year	First Year (2015)	Second Year (2016)	Third Year (2017)	
Salary (₹)	[1,80,000]	[1,80,000 + 10,000]		

$$a = 1,80,000 \quad d = 10,000 \quad n = ? \quad t_n = 2,50,000 \text{ ₹}$$

$$t_n = a + (n - 1)d$$

$$2,50,000 = 1,80,000 + (n - 1) \times 10,000$$

$$(n - 1) \times 10000 = 70,000$$

$$(n - 1) = 7$$

$$n = 8$$

In the 8th year her annual salary will be ₹ 2,50,000.





Class - X

2023-24

Board: Maharashtra

FINANCIAL PLANNING

1. INTRODUCTION:

- Taxes that existed before were Excise Duty, Custom Duty, VAT, Entertainment tax, Central sales tax, Service tax, Octroi etc.

All these taxes are subsumed under GST, that is why GST is One nation, One tax, One market. GST is in effect from 1st of July 2017.

2. TAX INVOICE:

Tax Invoice of goods purchase (Sample)										
SUPPLIER: A to Z SWEET MART						GSTIN: 27ABCDE1234HIZ5				
143, Shivaji Rasta, Mumbai: 400001, Maharashtra.										
Mo No. 9263692111 e-mail - atoz@gmail.com										
Invoice No. GST/110						Invoice Date: 31-Jul-2017				
S. No.	HSN code	Name of Product	Rat	Quantity	Taxable Amount	CGST		SGST		Total ₹.
						Rate	Tax	Rate	Tax	
1	210690	Pedhe	₹ 400 per kg.	500 gm.	200.00	2.5%	5.00	2.5%	5.00	210.00
2	210691	Chocolate	₹ 80	1Bar	80.00	14%	11.20	14%	11.20	102.40
3	2105	Ice-cream	₹ 200 per pack	1 pack (500 gm.)	200.00	9%	18.00	9%	18.00	236.00
4	1905	Bread	₹ 35	1 pack	35.00	0%	0.00	0%	0.00	35.00
5	210690	Butter	₹ 500 per kg.	250 gm	125.00	6%	7.50	6%	7.50	140.00
Total Rupees							41.70		41.70	723.40

- Let us understand the meaning of some new words given in this bill.

CGST - Central Goods and Service Tax.

SGST - State Goods and Service Tax.

GSTIN - Goods and Service Tax Identification Number.

HSN - Harmonized System of Nomenclature.

SAC - Service Accounting Code.

Now, CGST, i.e. Central Goods and Service Tax is to be paid to the Central Government whereas SGST, i.e. State Goods and Service Tax is to be paid to State Government. It is compulsory to get GSTIN, if the turn-over of a dealer in previous financial year exceeds ₹20 lacs. GSTIN includes 15 alpha numerals.

For ex.

e.g. : 27 A B C D E 1 2 3 4 H 1 Z 5

10 digit PAN of the firm

1 represents one registration

Uniform of all (By default)

2 digits for state code

Check sum digit (confirms the validity of GSTIN when entered on GST portal)

Note: Here 27 is the state code of Maharashtra. From 27, one can understand that a person or a firm is registered in Maharashtra.



Tax Invoice of Service provided (Sample)								
Food Junction, Shivpur, Pune					Invoice No. :58			
Mob. No. 7588580000					E-mail: ahar.khed@yahoo.com			
GSTIN: 27AAAA5555B1ZA					Invoice Date: 25 Dec., 2017			
SAC	Food Items	Qty.	Rate (in ₹)	Taxable amount	CGST		SGST	
9961	Coffee	1	20	20.00	2.5%	₹ 0.50	2.5%	₹ 0.50
9963	Masala Tea	1	10	10.00	2.5%	₹ 0.25	2.5%	₹ 0.50
9962	Masala Dosa	1	60	120	2.5%	₹ 3	2.5%	₹ 3
Total				150		₹ 3.75		₹ 3.75
Grand Total = ₹ 157.50								

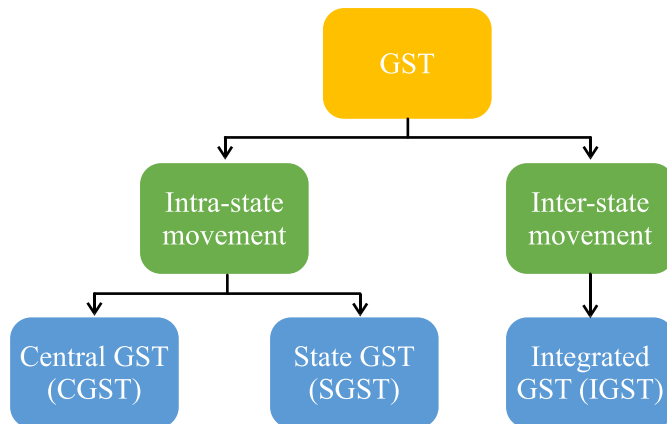
Sr. No	Types	Rate of GST	Goods and services items list
I	Zero rated	0%	Goods - Essential Commodities like food grains, fruits, vegetables, milk, salt, earthen pots etc. Services - Charitable trust activities, transport of water, use of roads and bridges, public library, agriculture related services, Education and Health care services etc.
II	Low rated	5%	Goods- Commonly used items- LPG cylinder, Tea, coffee, oil, Honey, Frozen vegetables, spices, sweets etc. Services - Railway transport services, bus transport services, taxi services, Air transport (economy class), Hotels providing food and beverages etc.
III	Standard rated (I slab)	12%	Goods- Consumer goods: Butter, Ghee, Dry fruits, Jam, Jelly, Sauces, Pickles. Mobile phone etc. Services - Printing job work, Guest house, Services related to construction business.
IV	Standard rated (II slab)	18% (Most of the Goods and services are included)	Goods - Marble, Granite, Perfumes, Metal items, Computer, Printer, Monitor, CCTV etc. Services - Courier services, Outdoor catering, Circus, Drama, Cinema, Exhibitions, Currency exchange, Broker Services in share trading etc.
V	Highly rated	28%	Goods - Luxury items, Motor Cycles and spare parts, Luxury cars, Pan-masala, Vacuum cleaner, Dish washer, AC, Washing machine, Fridge, Tobacco products, Aerated water etc. Services – Five-star Hotel accommodation Amusement parks, Water parks, Theme parks, Casino, Race course, IPL games, Air transport (business class) etc.

Note: - The rates and types of GST are as prescribed by the government at the time of writing this chapter. GST rates are subject to change. Electricity, petrol, diesel etc are not under purview of GST.

Note: Rates on goods and services, SAC and HSN codes are only for information, no need to remember them.



Taxes Under GST



Note: Value of Goods on which GST is levied is called taxable value. Total value or Invoice value is the value with GST. If not mentioned take the selling prices as taxable price. Remember that in tax invoice CGST amount is always equal to SGST amount.

Formulae:

1. $CGST = SGST$
2. $GST = CGST + SGST$
3. $CGST = SGST = \frac{1}{2} GST$
4. Amount payable by customer = Taxable value + GST

Example:

Courier service agent charged total ₹ 590 to courier a parcel from Nashik to Nagpur. In the tax invoice taxable value is ₹ 500 on which CGST is ₹ 45 and SGST is ₹ 45. Find the rate of GST charged for this service.

Solution:

Total GST = CGST + SGST = 45 + 45 = ₹ 90.

$$\text{Rate of GST} = \frac{90}{500} \times 100 = 18\%$$

∴ Rate of GST charged by agent is 18%.

Example:

Shreekar bought a Laptop with 10% discount on printed price. The printed price of that Laptop was ₹ 50,000. 18% GST was charged on discounted price. Find the amount of CGST and SGST. What amount did Shreekar pay?

Solution:

Discount = 10% of 50,000 = ₹ 5,000

∴ Taxable value of Laptop = 50,000 – 5,000 = ₹ 45,000.

∴ Rate of GST = 18% (Given) ∴ Rate of CGST = 9%

$$\therefore CGST = 9\% \text{ of } 45,000 = \frac{9}{100} \times 45000 = ₹ 4050.$$

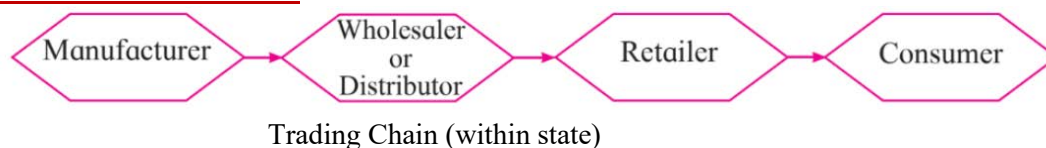
∴ SGST = ₹ 4050.

∴ Amount paid = 45000 + 4050 + 4050 = ₹ 53,100.

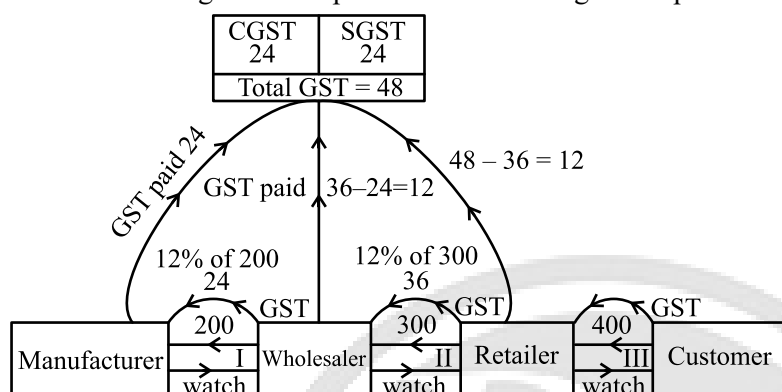
Ans. Shreekar paid ₹ 53,100 for the Laptop.



3. GST IN TRADING CHAIN:



- Let's learn through an example how GST is charged and paid to the government at every stage of trading.



- Here all the three financial transactions took place in one state. Therefore three tax invoices were generated as follows. Each tax invoice shows the brief computation of GST.

GST in Tax Invoice I	GST in Tax Invoice II	GST in Tax Invoice III
Price of watch = ₹ 200	Price of watch = ₹ 300	Price of watch = ₹ 400
CGST 6% = ₹ 12	CGST 6% = ₹ 18	CGST 6% = ₹ 24
SGST 6% = ₹ 12	SGST 6% = ₹ 18	SGST 6% = ₹ 24
Total = ₹ 224	Total = ₹ 336	Total = ₹ 448
Tax invoice of manufacturer B2B	Tax invoice of wholesaler B2B	Tax invoice of retailer B2C

- Trading between GSTIN holders is known as Business to Business, in short B2B. Trading between GSTIN holder and consumer is known as Business to Consumer, in short B2C. This is the last link in the trading chain.

Input Tax: When trader Pay GST at the time of purchase is called Input Tax.

Output Tax: When trader collects GST at the time of sale is called as Output tax.

Input Tax Credit: At the time of paying GST to the government a trader deducts the input tax from the output tax and pays the remaining tax. This deduction of input tax is called **Input Tax Credit**.

GST Payable = output tax – Input tax

Example:

Mr. Rohit is a retailer. He paid GST of ₹ 6500 at the time of purchase. He collected GST of ₹ 8000 at the time of sale. (i) Find his input tax and out-put tax. (ii) What is his Input tax credit? (iii) Find his payable GST. (iv) Hence find the payable CGST and payable SGST.

Solution:

Mr. Rohit's payable GST means, GST to be paid to the government by Mr. Rohit.

(i) Output tax (tax collected at the time of sale) = ₹ 8000.

(ii) Input tax (tax paid at the time of purchase) = ₹ 6500

∴ ITC = ₹ 6500.

(iii) GST payable = Output tax – ITC

= ₹ 8000 – ₹ 6500 = ₹ 1500

(iv) ∴ payable CGST = 1500/2 = ₹ 750 and payable SGST = ₹ 750.

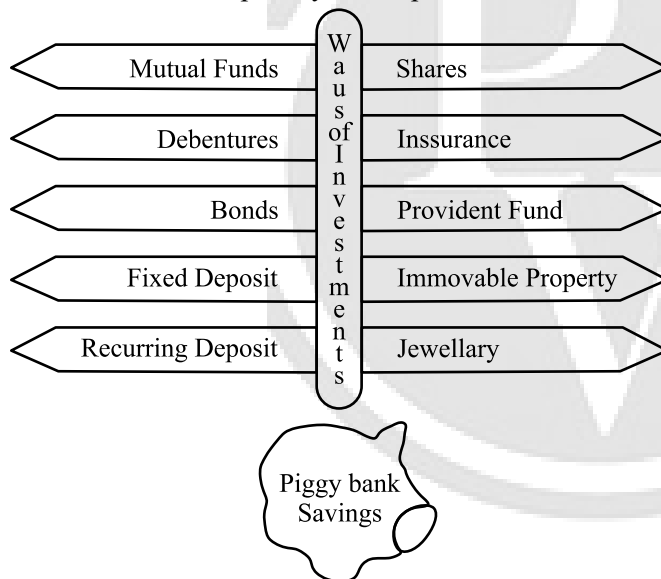


Features of GST

- Many Indirect Taxes are subsumed under GST.
- No dispute between Goods and Services.
- Statewise Registration for traders.
- GSTIN holder needs to keep all the records and should pay GST in time.
- Transparency in transactions.
- This tax system is simple and easy to understand.
- Removal of cascading effect of taxes hence the prices are controlled
- Increase in Quality of Goods and Services as they are globally competitive.
- Boost to 'Make in India' project.
- Technology driven tax system leads to speedy decisions.
- Goods and Service Tax system is a Dual model, as equal amount of tax is levied by Central and State governments.

4. SAVINGS AND INVESTMENTS:

- In the previous class we have learnt the importance of savings and investments, which you might have started practising whenever possible. We develop good habits to maintain physical health, in the same way we should develop a habit of saving and investing regularly to maintain financial health. There are many different ways of investments. So deep study and experience both are essential.



Share Holder: A person who owns the share is called a share holder. The shareholder is a part owner of the company in the proportion of number of shares he/she holds.

Stock Exchange: It is a place where buying and selling of shares take place. It is also known as share market or stock market, equity market or capital market. Companies should be listed in the stock market for trading.

Face Value (FV): The value printed on the share certificate is called the Face value of the share. It is also called Nominal value or Printed value or par value.

Market Value (MV): The price at which the shares are sold or purchased in the stock market is called Market value (MV) of the share.

Dividend: The part of annual profit of a company which is distributed per share among shareholders is called dividend. If the company is performing well then the value of share capital increases hence the price of the share goes up. As a result company gives good dividend. For the shareholders the dividend income is tax free.



5. COMPARISON OF FV AND MV:

- (1) If $MV > FV$ then the share is at premium.
- (2) If $MV = FV$ then the share is at par.
- (3) If $MV < FV$ then the share is at discount.

For example:

- (1) suppose $FV = ₹10$, $MV = ₹15$ and $15 - 10 = ₹5$
 $\therefore$ The share is at premium of ₹5, as $MV > FV$
- (2) suppose $FV = ₹10$, $MV = ₹10$ and $10 - 10 = 0$
 $\therefore$ The share is at par. As $MV = FV$
- (3) suppose $FV = ₹10$, $MV = ₹7$ and $10 - 7 = 3$
 $\therefore$ The share is at discount. As $MV < FV$.

Formulae:

- (1) Amount Invested = $M.V. \times \text{NO of shares}$
- (2) Dividend per share = $\text{Rate of dividend} \times F.V$
- (3) Dividend Income = $\text{Rate of dividend} \times F.V \times \text{No. of shares}$
- (4) Rate of return = $\frac{\text{Dividend Income}}{\text{Amount Invested}} \times 100$

Example:

If 50 shares of $FV ₹100$ each are purchased for $MV ₹120$. Find the sum invested.

Solution:

$$\text{Sum invested} = \text{number of shares} \times MV = 50 \times 120 = ₹6000$$

Example:

If 50 shares of $FV ₹10$ were purchased for MV of ₹25. Company declared 30% dividend on the shares then find (1) Sum investment (2) Dividend received (3) Rate of return.

Solution:

$FV = ₹10$, $MV = ₹25$, Number of shares = 50.

(1) $\therefore$ Sum investment = $25 \times 50 = ₹1250$.

(2) Dividend per share = $10 \times \frac{30}{100} = ₹3$

$\therefore$ Total dividend received = $50 \times 3 = ₹150$.

(3) Rate of return = $\frac{\text{Dividend income}}{\text{Sum invested}} \times 100$

$$= \frac{150}{1250} \times 100 = 12\%$$

Ans: (1) Sum invested is ₹1250

(2) Dividend received is ₹150

(3) Rate of return is 12%.

6. BROKERAGE AND TAXES ON SHARE TRADING:

- **Brokerage:** We directly can't go to the stock market and buy or sell shares, only the registered members or organization (agency) of the stock market can buy or sell on our behalf. These members are called '**Share Brokers**'.
- For catering the service of buying and selling of shares they charge some amount which is called '**Brokerage**'.
- Brokerage is paid on the Market value of the share.
Situation (I) At the time of buying shares:
- Buying price of 1 share = $MV + \text{Brokerage} + \text{GST on brokerage}$ (If mentioned)
- Situation (II) At the time of selling shares.**
- Selling price per share = $MV - \text{Brokerage} - \text{GST on brokerage}$ (If mentioned)
- Rate of GST is 18% on brokerage.



Example:

Bashirkhan purchased 100 shares of MV ₹ 40. Brokerage paid at the rate of 0.5% and rate of GST on brokerage is 18%. Find the total amount he paid for the share purchase.

Solution:

Value of 100 shares = $40 \times 100 = ₹ 4000$.

Brokerage per share = $\frac{0.5}{100} \times 40 = ₹ 0.20$.

∴ Cost of one share = MV + Brokerage
= $40 + 0.20 = ₹ 40.20$.

∴ Cost of 100 shares = $40.20 \times 100 = ₹ 4020$

∴ Brokerage on 100 share = $0.20 \times 100 = ₹ 20$

∴ GST = $\frac{18}{100} \times 20$
= ₹ 3.60.

Ans. Bashirkhan paid ₹ 4020 + ₹ 3.60 = ₹ 4023.60 for 100 shares.

7. MUTUAL FUND - MF:

- We have learnt that a group of persons come together to form a company. They raise capital from the society by issuing shares.
 - If company performs well, then the investors of the company get benefits in terms of dividend, bonus shares and increase in the market value gives more profit on investments.
 - Sometimes it might happen that instead of profit an investor may incur a loss. Can we reduce this loss? Is there a way to reduce the risk in investments? Yes, to overcome this more people invest in Mutual Funds.
 - In Mutual Fund, many investors with common objectives give their money to the professional experts.
 - They not only invest in one type of shares but also invest in various other schemes.
 - As a result, investment is diversified which reduces risk factor and total dividend or profit is divided equally among the investors.
 - Mutual Fund is a professionally managed investment scheme, usually run by an AMC i.e. Asset Management Company.
 - They invest the money given by the investors in different schemes e.g. equity fund (in shares), debt fund (in debentures, bonds etc.) or balanced funds as per the investor's choice.
 - As we get 'shares' for the investment in share-market, we get '**units**' when we invest in mutual fund.
 - The market value of 'a unit' is called '**NAV**' (Net Asset Value)
 - NAV of one unit $\times$ Number of units = Total fund value.
- Note:** As the market value of share changes frequently NAV of a unit also changes. One can redeem the units when needed.
- Investments in FDs of nationalised bank or Indian Postal services are more-secured and safe, in comparison with other investments, but the rate of return is low.
 - It hardly helps to overcome the rate of inflation.
 - One must remember always that if the money is invested wisely it generates more money.
 - For this the knowledge of financial planning is of great help.
 - Investments in shares and mutual funds should be made carefully because risk and returns always go hand in hand).
 - So the habit of regular and deep study is the only key.



8. **SYSTEMATIC INVESTMENT PLAN:**

- Suppose, one does not want to invest a big amount at once, then one could invest small amounts at regular time intervals e.g. ₹ 500 per month could be invested in mutual fund.
- Investment could be done monthly or quarterly.
- This way of investment is called SIP.
- SIP develops discipline of savings.
- SIP is a good option which in long term can achieve one's financial goals.
- Investment in mutual funds through SIP for a long term is beneficial.
- It protects investor from market fluctuations.
- One should invest in mutual fund for minimum of 3 to 5 years to get better returns and it is best if investment is for 10 to 15 years.

Benefits of Mutual Funds

- Professional fund managers.
- Diversifications of funds.
- Transparency and sufficiently safe investment.
- Liquidity - redemption of units can be done.
- Limited risks.
- Advantage of long term and short term gain.
- Investments in funds like ELSS are admissible for deduction under section 80C of income tax.

Example:

If the total value of the mutual fund scheme is ₹ 200 crores and 8 crore units are issued then find the NAV of one unit.

Solution:

$\text{NAV} = ₹ 200 \text{ crore} / 8 \text{ crore units} = ₹ 25 \text{ per unit.}$

Example:

If NAV of one unit is ₹ 25, then how many units will be allotted for the investment of ₹ 10,000?

Solution:

$\text{Number of units} = \text{sum invested} / \text{NAV} = 10,000/25 = 400 \text{ units.}$





Class - X

2023-24

Board: Maharashtra

PROBABILITY

1. PROBABILITY:

- Probability is a measure of the likelihood of an event to occur.
- Probability can range from 0 to 1, where 0 means the event to be an impossible one and 1 indicates a certain event.
- The probability of all the events in a sample space adds up to 1.

2. RANDOM EXPERIMENT:

- The experiment in which all possible results are known in advance but none of them can be predicted with certainty and there is equal possibility for each result is known as a 'Random experiment'.
- For example, Tossing a coin, throwing a die, picking a card from a set of cards bearing numbers from 1 to 50, picking a card from a pack of well shuffled playing cards, etc.

3. OUTCOME:

- Result of a random experiment is known as an 'Outcome'.
- Ex. In a random experiment of tossing a coin - there are only two outcomes i.e. Head (H) or Tail (T)
- In a random experiment of throwing a die, there are 6 outcomes, according to the number of dots on the six faces of the die. 1 or 2 or 3 or 4 or 5 or 6.

4. EQUALLY LIKELY OUTCOMES:

- If a die is thrown, any of the numbers from 1, 2, 3, 4, 5, 6 may appear on the upper face. It means that each number is equally likely to occur. However, if a die is so formed that a particular face come up most often, then that die is biased. In this case the outcomes are not likely to occur equally.
- Here, we assume that objects used for random experiments are fair or unbiased.
- A given number of outcomes are said to be equally likely if none of them occurs in preference to others.
- For example if a coin is tossed, possibilities of getting head or tail are equal.

5. SAMPLE SPACE:

- The set of all possible outcomes of a random experiment is called the sample space.
- It is denoted by ' S ' or ' Ω ' (A greek letter 'Omega').
- Each element of sample space is called a 'sample point'.
- The number of elements in the set ' S ' is denoted by $n(S)$.
- If $n(S)$ is finite, then the sample space is said to be a finite sample space.
Following are some examples of finite sample spaces.



S. No.	Random experiment	Sample space	Number of sample points in S
1	One coin is tossed	$S = \{H, T\}$	$n(S) = 2$
2	Two coins are tossed	$S = \{HH, HT, TH, TT\}$	$n(S) = 4$
3	Three coins are tossed	$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$	$n(S) = 8$
4	A die is thrown	$S = \{1, 2, 3, 4, 5, 6\}$	$n(S) = 6$
5	Two dice are thrown	$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$	$n(S) = 36$
6	A card is drawn from a pack bearing numbers from 1 to 25	$S = \{1, 2, 3, 4, \dots, 25\}$	$n(S) = 25$
7	A card is drawn from a well-shuffled pack of 52 playing cards.	Diamond: Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King Spade: Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King Heart: Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King Club: Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King	$n(S) = 52$

6. **EVENT:**

- The outcomes satisfying particular condition are called favourable outcomes.
- A set of favourable outcomes of a given sample space is an 'event'.
- Event is a subset of the sample space.
- Events are generally denoted by capital letters A, B, C, D etc.
- For example, if two coins are tossed and A is the event of getting at least one tail, then the favourable outcomes are as follows. $A = \{TT, TH, HT\}$
- The number of elements in the event A is denoted by $n(A)$. Here $n(A) = 3$.

Types of Events

- (a) **Certain Event/Sure event:** An event is known as the Certain Event when it contains all the sample points of sample space.

e.g. Random experiment: A die is thrown. Let A be the event of getting a score on the upper face which is less than 7.

Here, $S = \{1, 2, 3, 4, 5, 6\}$ and $A = \{1, 2, 3, 4, 5, 6\}$

Here, event A has all the elements of S, therefore event A is a certain event.



- (b) Impossible Event: An event is called an Impossible Event if it does not contain any sample point of sample space.
e.g. Random experiment: A die is thrown. Let A be the event of getting a score on the upper face which is divisible by 7.
Here, $S = \{1, 2, 3, 4, 5, 6\}$ and $A = \{ \}$.
Event A is an Impossible event if $A = \phi$ or $A = \{ \}$.
- (c) Simple/Elementary Event: An event consisting of only one sample point of a sample space is called an elementary event.
e.g. Random experiment: A die is thrown. Let A be the event of getting a score on the upper face which is divisible by 5.
Here $S = \{1, 2, 3, 4, 5, 6\}$ and $A = \{5\}$
 $\therefore n(A) = 1$
 $\therefore$ Event A is an elementary event.
- (d) Mutually exclusive Events: Two events are called Mutually exclusive Events if they do not have any sample points in common.

Example:

If $A = \{1, 2, 3\}$ and $B = \{4, 5\}$ are two events then, A and B are Mutually exclusive events, since they do not have any sample point in common.

Mathematically: $A \cap B = \phi$ (Empty set)

- (e) Exhaustive Event: Two events are said to be exhaustive events if their union is sample space.
e.g. Let S be the sample space
 $S = \{1, 2, 3, 4, 5, 6\}$
 $A = \{1, 2, 3\}$
 $B = \{3, 4, 5, 6\}$
 $A \cup B = \{1, 2, 3, 4, 5, 6\}$ which is sample space.
 $\therefore A \cup B = S$

Example:

Two coins are tossed simultaneously. Write the sample space (S) and number of sample points $n(S)$. Also write the following events in the set form and write the number of sample points in each event.

- (i) Condition for event A : to get at least one tail.
(ii) Condition for event B : to get only one head.
(iii) Condition for event C : to get at most one tail.
(iv) Condition for event D : to get no head.

Solution:

If two coins are tossed simultaneously,
 $S = \{HH, HT, TH, TT\}$ $n(S) = 4$

- (i) Condition for event A : at least one head.
 $A = \{HH, HT, TH\}$ $n(A) = 3$
(ii) Condition for event B : only one head.
 $B = \{HT, TH\}$ $n(B) = 2$
(iii) Condition for event C : at most one tail.
 $C = \{HH, HT, TH\}$ $n(C) = 3$
(iv) Condition for event D : No head.
 $D = \{TT\}$ $n(D) = 1$



Example:

A bag contains 50 cards. Each card bears only one number from 1 to 50. One card is drawn at random from the bag. Write the sample space. Also write the events A , B and find the number of sample points in them.

- (i) Condition for event A : the number on the card is divisible by 6.
- (ii) Condition for event B : the number on the card is a complete square.

Solution:

$$S = \{1, 2, 3, \dots, 49, 50\}, n(S) = 50$$

- (i) Condition for event A : number is divisible by 6 .
 $A = \{6, 12, 18, 24, 30, 36, 42, 48\}$ $n(A) = 8$
- (ii) Condition for event B : the number on the card is a complete square.
 $B = \{1, 4, 9, 16, 25, 36, 49\}$ $n(B) = 7$

Example:

A sanitation committee of 2 members is to be formed from 3 boys and 2 girls. Write sample space ' S ' and number of sample points $n(S)$. Also write the following events in set form and number of sample points in the event.

- (i) Condition for event A : at least one girl must be a member of the committee.
- (ii) Condition for event B : Committee must be of one boy and one girl.
- (iii) Condition for event C : Committee must be of boys only.
- (iv) Condition for event D : At the most one girl should be a member of the committee.

Solution:

Suppose B_1, B_2, B_3 are three boys and G_1, G_2 are two girls

Out of these boys and girls, a sanitation committee of two members is to be formed.

$$\therefore S = \{B_1B_2, B_1B_3, B_2B_3, B_1G_1, B_1G_2, B_2G_1, B_2G_2, B_3G_1, B_3G_2, G_1G_2\} \therefore n(S) = 10$$

- (i) Condition for event A is that at least one girl should be in the committee.
 $A = \{B_1G_1, B_1G_2, B_2G_1, B_2G_2, B_3G_1, B_3G_2, G_1G_2\} \therefore n(A) = 7$
- (ii) Condition for event B is that one boy and one girl should be there in the committee.
 $B = \{B_1G_1, B_1G_2, B_2G_1, B_2G_2, B_3G_1, B_3G_2\} \therefore n(B) = 6$
- (iii) Condition for event C is that there should be only boys in the committee.
 $C = \{B_1B_2, B_1B_3, B_2B_3\} \therefore n(C) = 3$
- (iv) Condition for event D is that there can be at most one girl in the committee.
 $D = \{B_1B_2, B_1B_3, B_2B_3, B_1G_1, B_1G_2, B_2G_1, B_2G_2, B_3G_1, B_3G_2\} \therefore n(D) = 9$

Example:

Two dice are rolled, write the sample space ' S ' and number of sample points $n(S)$. Also write events and number of sample points in the event according to the

Solution: Sample space,

$$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\} \therefore n(S) = 36$$

- (i) Event E : The sum of the digits on the upper face is a prime number.
 $E = \{(1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4, 3), (5, 2), (5, 6), (6, 1), (6, 5)\}$
 $\therefore n(E) = 15$



- (ii) Event F : The sum of the digits on the upper face is a multiple of 5.

$$F = \{(1, 4), (2, 3), (3, 2), (4, 1), (4, 6), (5, 5), (6, 4)\} \therefore n(F) = 7$$

- (iii) Event G : The sum of the digits on the upper face is 25.

$$G = \{\} = \phi \quad \therefore n(G) = 0$$

- (iv) Event H : The number on upper face of first die is less than the digit on second die.

$$H = \{(1, 2) (1, 3) (1, 4) (1, 5) (1, 6) (2, 3) (2, 4) (2, 5) (2, 6) (3, 4) (3, 5) (3, 6) (4, 5) (4, 6) (5, 6)\} \therefore n(H) = 15$$

7. PROBABILITY OF AN EVENT:

- In Mathematical language, when possibility of an expected event is expressed in number, it is called 'Probability'.
- It is expressed as a fraction or percentage using the following formula.
- For a random experiment, if sample space is 'S' and 'A' is an expected event then probability of 'A' is $P(A)$. It is given by following formula.

$$P(A) = \frac{\text{Number of sample points in event } A}{\text{Number of sample points in sample spaces}} = \frac{n(A)}{n(S)}$$

The probability of any event is from 0 to 1 or 0% to 100%.

If E is any event, $0 \leq P(E) \leq 1$ or $0\% \leq P(E) \leq 100\%$.

Example:

Find the probability of the following, when one coin is tossed.

- (i) getting head
- (ii) getting tail

Solution:

Let 'S' be the sample space.

$$S = \{H, T\} \quad n(S) = 2$$

- (i) Let event A be getting head

$$A = \{H\} \quad \therefore n(A) = 1$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{1}{2}$$

- (ii) Let event B be getting tail

$$B = \{T\} \quad \therefore n(B) = 1$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{1}{2}$$

Example:

If one die is rolled then find the probability of each of the following events.

- (i) Number on the upper face is prime
- (ii) Number on the upper face is even.

Solution:

'S' is the sample space.

$$S = \{1, 2, 3, 4, 5, 6\} \quad \therefore n(S) = 6$$

- (i) Event A : Prime number on the upper face.

$$A = \{2, 3, 5\} \quad \therefore n(A) = 3$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$\therefore P(A) = \frac{3}{6} = \frac{1}{2}$$



- (ii) Event B : Even number on the upper face.

$$B = \{2, 4, 6\} \quad \therefore n(B) = 3$$

$$P(B) = \frac{n(B)}{n(S)}$$

$$\therefore P(B) = \frac{3}{6} = \frac{1}{2}$$

Example:

A card is drawn from a well shuffled pack of 52 playing cards. Find the probability of each event. The card drawn is

- (i) a red card
(ii) a face card

Solution:

'S' is the sample space. $\therefore n(S) = 52$

Event A : Card drawn is a red card.

Total red cards = 13 hearts + 13 diamonds = 26

$$\therefore n(A) = 26$$

$$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{26}{52} = \frac{1}{2}$$

Event B : Card drawn is a face card.

Total face cards = 12 $\therefore n(B) = 12$

$$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{12}{52} = \frac{3}{13}$$

Example:

A box contains 5 strawberry chocolates, 6 coffee chocolates and 2 peppermint chocolates. Find the probability of each of the following events, if one of the chocolates is picked from the box at random.

- (i) it is a coffee chocolate.
(ii) it is a peppermint chocolate.

Solution:

Sample space is 'S' and $n(S) = 5 + 6 + 2 = 13$

Event A : it is a coffee chocolate

$$\therefore n(A) = 6$$

$$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{6}{13}$$

Event B : it is a peppermint chocolate

$$\therefore n(B) = 2$$

$$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{2}{13}$$





Class - X

2023-24

Board: Maharashtra

STATISTICS

1. STATISTICS:

- Statistics is something that surrounds us every day - we are constantly bombarded with statistics, in the form of polls, tests, ratings etc. It is a branch of Mathematics which deals with the collection, presentation and analysis of numerical data and drawing the conclusions on the basis of the same.
- Statistics is used to study the problems in Biology, Psychology, Economics, Education, Sociology, Trade etc. It is widely used in industries to control the quality of products.
- In our X^{th} standard syllabus, we have four things to learn:
 - (A) Measures of Central Tendency of Grouped Data.
 - (B) Pie-Diagram.
 - (C) Histogram.
 - (D) Frequency Polygon or Frequency Curve

2. MEASURES OF CENTRAL TENDENCY OF GROUPED DATA:

- The most basic concept in statistics is the idea of an average (arithmetic mean). An average is a single number which represents the idea of a typical value. There are three different numbers which can represent the idea of an average value. The three values are the mean, the median and the mode.
- **Mean:** Given a set of values, the mean is sum of all the values, divided by the total number of values. Mean summarizes the properties of entire data.
There are three different methods to find Mean for the given data.

3. METHOD-I: DIRECT METHOD:

- In this method, table will have four columns.

Step-1: In 1st column, write the given Classes.

Step-2: In 2nd column, write the Classmark/Midvalues (x_i) for each of the class.

Find the class mark of the first class, the remaining class marks could be found by adding class width (h) to the previous class mark.

Step-3: In 3rd column, write the Frequency (f_i) of all the classes. Find the sum of all frequencies i.e. Σf_i .

Step-4: In 4th column, find the product of Classmark/Midvalues (x_i) and frequency (f_i), i.e. $f_i x_i$, for each of the classes. Find the sum of all ($f_i x_i$) values i.e. $\Sigma f_i x_i$

Step-5: Calculate, Mean ($\bar{x}$) =
$$\frac{\Sigma f_i x_i}{\Sigma f_i}$$



Example:

The percentage of marks of 50 students in a test is given in the following table. Find the mean of the percentage.

Percentage of marks	0 – 20	20 - 40	40 - 60	60 - 80	80 - 100
No. of students	3	7	15	20	5

Solution:

The following table is prepared as per steps.

Class (Percentage of marks)	Class mark x_i	Frequency (No. of students) f_i	Class mark $\times$ frequency $x_i f_i$	$\bar{X} = \frac{\sum x_i f_i}{\sum f_i}$ $= \frac{2840}{50}$ $= 56.8$ $\therefore$ The mean the percentage $= 56.8$
0 – 20	10	3	30	
20 – 40	30	7	210	
40 – 60	50	15	750	
60 – 80	70	20	1400	
80 – 100	90	5	450	
Total		$N = \sum f_i = 50$	$\sum x_i f_i = 2840$	

4. **METHOD-II (Assumed Mean Method)**

Assumed Mean Method or shift of origin method:

In this method, table will have five columns.

Step-1: In 1st column, write the given classes.

Step-2: In 2nd column, write the Classmark/Mid values (x_i) for each of the class.

Step-3: Select any value in the Classmark (x_i) column as Assumed mean. It is denoted by A. Generally, middle value of the column is selected.

Step-4: In 3rd column, find deviations using formula $d_i = x_i - A$ for first class and the remaining d_i could be found by adding class width to the previous d_i .

Step-5: In 4th column, write the Frequency (f_i) of all the classes. Also, find the sum of all frequencies, i.e. $\sum f_i$

Step-6: In 5th column, find the product of deviations (d_i) and frequency (f_i), i.e. $f_i d_i$ for each of the classes. Find the sum of all ($f_i d_i$) values i.e. $\sum f_i d_i$

Step-7: Calculate $\bar{d} = \frac{\sum f_i d_i}{\sum f_i}$

Step-8: Calculate Mean $\bar{x} = A + \bar{d}$



Example:

The following table shows the frequency table of daily wages of 50 workers in a trading company. Find the mean wages of a worker, by assumed mean method.

Daily Wages (Rs)	200 – 240	240 – 280	280 – 320	320 – 360	360 – 400
Frequency (No. of workers)	5	10	15	12	8

Solution:

Let us take the assumed mean $A = 300$.

Class (₹ Wage)	Class mark x_i	$d_i = x_i - A$ $d_i = x_i - 300$	Frequency (No. of workers) f_i	Frequency × Deviation $f_i d_i$
200 – 240	220	– 80	5	– 400
240 – 280	260	– 40	10	– 400
280 – 320	300 → A	0	15	0
320 – 360	340	40	12	480
360 – 400	380	80	8	640
Total			$\sum f_i = 50$	$\sum f_i d_i = 320$

$$\bar{d} = \frac{\sum f_i d_i}{\sum f_i} = \frac{320}{50} = 6.4$$

$$\begin{aligned}\text{Mean, } \bar{X} &= A + \bar{d} \\ &= 300 + 6.4 \\ &= 306.40\end{aligned}$$

The mean of daily wages = 306.40 ₹

5. METHOD-III: (Step Deviation Method)

- Step Deviation Method or shift of origin and scale method.

Width of class (g) = Difference between any two consecutive upper limits or lower limits. In this method, table will have 6 columns.

Step-1: In 1st column, write the given classes.

Step-2: In 2nd column, write the Class mark/ Mid-values (x_i) for each of the class.

Step-3: In 3rd column, write the values of the d_i , where $d_i = x_i - A$

Step-4: In 4th column, write the value of the u_i where $u_i = \frac{x_i - A}{g} = \frac{d_i}{g}$

Step-5: In 5th column, write the frequencies f_i and find its sum.

Step-6: In 6th column, find the product of u_i and frequency, f_i i.e. $f_i u_i$ for each of the classes. Find the sum of all ($f_i u_i$) values. i.e. $\sum f_i u_i$

Step-7: Find mean of u_i using the formula $\bar{u} = \frac{\sum f_i u_i}{\sum f_i}$

Step-8: Calculate, Mean ($\bar{X}$) = $A + g\bar{u}$



Example:

The following table shows the funds collected by 50 students for flood affected people. Find the mean of the funds.

Fund (Rupees)	0 -500	500 -1000	1000 -1500	1500 -2000	2000 -2500	2500 -3000
No. of students	2	4	24	18	1	1

If the number of scores in two consecutive classes is very low, it is convenient to club them. So, in the above example, we club the classes 0 – 500, 500 – 1000 and 2000 – 2500, 2500 – 3000. Now the new table is as follows.

Fund (Rupees)	0 – 1000	1000 – 1500	1500 – 2000	2000 – 3000
No. of students	6	24	18	2

Solution:

Let $A=1250$, examining all d_i , $g = 250$.

Class Fund (₹)	Class mark x_i	$d_i = x_i - A$ $= x_i - 1250$	$u_i = \frac{d_i}{g}$	Frequency f_i	$f_i u_i$
0 – 1000	500	- 750	- 3	6	- 18
1000 – 1500	1250 $\rightarrow A$	0	0	24	0
1500 - 2000	1750	500	2	18	36
2000 – 3000	2500	1250	5	2	10
Total				$\sum f_i = 50$	$\sum f_i u_i = 28$

$$\bar{u} = \frac{\sum f_i u_i}{\sum f_i} = \frac{28}{50} = 0.56$$

$$\bar{u} g = 0.56 \times 250 = 140$$

$$\bar{X} = A + g \bar{u} = 1250 + 140 = 1390$$

$\therefore$ the average of the funds is ₹ 1390.

6. MEDIAN:

- Median is a measure of the 'Middle' value of the given data. When it is arranged. Median is often a better representative of a typical member of a group. If we take all values in a list and arrange them in increasing order, the number at the centre will be the median. Median gives you the central number of the group. Median is a number which divides data into two equal parts.

Steps to calculate Median

Step-1: Make classes continuous if they are not.

Step-2: In next column, write the frequencies.

Step-3: In next column, calculate the 'Cumulative frequency less than type'.

Step-4: Determine the median class, frequency of median class (f) and cumulative frequency of the class preceeding the median class ($c.f.$)



Step-5: Calculate Median $= L + \left(\frac{\frac{N}{2} - c.f.}{f} \right) h$

L = Lower limit of median class.

N = Total frequency.

$c.f.$ = Cumulative frequency of class preceeding the median class.

f = Frequency of median class.

h = class interval of the median class

- Steps to determine Median class

Step-1: Find the value of $N/2$, where N = Total frequency.

Step-2: Find the cumulative frequency less than type which is just greater than or equal to $N/2$. The corresponding class is a median class.

For Median,

- Classes have to be continuous.
- Cumulative frequency less than type has to be found for every sum.

Example:

Observe the following frequency distribution table. It shows the distances travelled by 60 public transport buses in a day. Find the median of the distance travelled.

Daily distance travelled (in Km)	200 – 209	210 – 219	220 – 229	230 – 239	240 – 249
No. of buses	4	14	26	10	6

Solution:

The classes in the table are not continuous.

The upper-class limit of a class and the lower class of its succeeding class differ by 1.

∴ Let us subtract $1 \div 2 = 0.5$ from the lower-class limit of each class and add to the upper-class limit of each class, and make the classes continuous.

Make a column of cumulative frequency 'less than' in the new table showing the continuous classes.

Given Class	Continuous classes	Frequency f_i	Cumulative frequency less than
200 – 209	199.5 – 209.5	4	4
210 – 219	209.5 – 219.5	14	18 $\rightarrow cf$
220 – 229	219.5 – 229.5	26 $\rightarrow f$	44
230 – 239	229.5 – 239.5	10	54
240 – 249	239.5 – 249.5	6	60

Here, total of frequencies $= \sum f_i = N = 60 \therefore \frac{N}{2} = 30$.

∴ 30th score is the approximate median.



First 18 scores are less than 219.5 and the remaining, $30 - 18 = 12$ scores are in the class 219.5 – 229.5. Therefore, 219.5 – 229.5 is the median class. The cumulative frequency of the class 219.5 – 229.5 is 44 .

In the formula,

L = Lower class limit = 219. 5, h = Class interval of the median class =10

$c f$ = The frequency of the class preceding the median class = 18,

f = The frequency of the median class = 26

$$\text{Median} = L + \left[\frac{\frac{N}{2} - cf}{f} \right] \times h$$

$$\therefore \text{Median} = 219.5 + \left(\frac{30/2 - 18}{26} \right) \times 10$$

$$= 219.5 + \left(\frac{12 \times 10}{26} \right)$$

$$= 219.50 + 4.62 = 224.12$$

$\therefore$ The median of the distance travelled is = 224.12 Km

7. **MODE:**

- Mode is the observation which occurs most frequently. The observation having maximum frequency is called mode. A data set has no mode when all the observations appearing in the data have the frequency 1. A data set has multiple modes when two or more values appear with the same highest frequency.

To find mode for the given data,

- Classes have to be continuous.
- If maximum frequency occurs twice in the data, then data will have two Modal classes.

Hence, Mode has to be found for both modal class.

Steps to calculate Mode:

Step-1: Make the classes continuous if they are not.

Step-2: Determine Modal class, Maximum frequency (f_1), frequency of class preceding modal class (f_0), frequency of class succeeding the modal class (f_2)

Step-3: Calculate,

$$\text{Mode} = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

L = Lower limit of modal class.

N = Total frequency.

f_1 = Maximum frequency.

f_0 = Frequency of class preceding modal class.

f_2 = Frequency of class succeeding the modal class.

h = class interval of the modal class.



Steps to determine Modal class:

Step-1: The class having maximum frequency is termed as Modal class.

Example:

The classification of children according to their ages, playing on a ground is shown in the following table. Find the mode of ages of the children.

Age-group of children (Yrs)	6 – 8	8 – 10	10 – 12	12 – 14	14 – 16
No. of children	43	58 $\rightarrow f_0$	70 $\rightarrow f_1$	42 $\rightarrow f_2$	27

From the table, we note that the maximum number of children is of the age-group 10 – 12. So the modal class is 10 – 12.

Solution:

Here $f_1 = 70$, and modal class is 10 – 12.

$\therefore$ in the given example,

L = Lower class limit of the modal class = 10

h = Class interval of the modal class = 2

f_1 = Frequency of the modal class = 70

f_0 = Frequency of the class preceding the modal class = 58

f_2 = Frequency of the class succeeding the modal class = 42

$$\text{Mode} = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

$$= 10 + \left[\frac{70 - 58}{2(70) - 58 - 42} \right] \times 2$$

$$= 10 + \left[\frac{12}{140 - 100} \right] \times 2$$

$$= 10 + \left[\frac{12}{40} \right] \times 2$$

$$= 10 + \frac{24}{40}$$

$$= 10 + 0.6$$

$$= 10.6$$

$\therefore$ the mode of the ages of children playing on the ground is 10.6 Years.

Pictorial Representation of Statistical Data:

- It is difficult to arrive at a conclusion with a raw data. So, it needs to be processed. Processing involves arranging data in order, classification, tabulation. Such a compact or condensed form of data are easy to understand. Further if it is graphically or diagrammatically represented, it becomes simple to understand and interpret.

The specific graphs associated with frequency distribution are as follows:

(i) Histogram (ii) Frequency polygon or Frequency curve (iii) Pie diagram

(i) Histogram:

Steps to draw histogram:

Step-1: Make the classes continuous if they are not continuous.

Step-2: Take continuous classes on X-axis with suitable scale. For writing the classes, we can either use 1 cm or 2 cm for each class.

Step-3: Take frequency on Y-axis with suitable scale.



Step-4: Draw rectangles on the X-axis for each class. The height of the rectangles is equal to the frequency of the respective class.

Note: Thus, the histogram is series of joint rectangles:

If there is a difference in the gap between the origin and lower limit of the first class and class width, then a '——' krink mark is drawn near the origin on the X-axis. The mark can be used on Y-axis to draw a graph of optimum site.

Example:

The table below shows the net asset value (NAV) per unit of mutual funds of some companies.

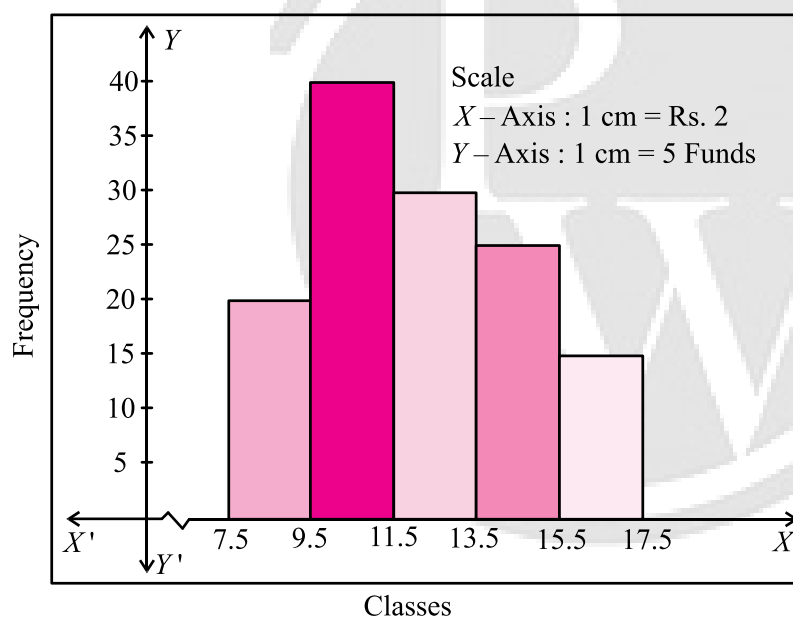
Draw a histogram representing the information.

NAV (₹)	8 – 9	10 – 11	12 – 13	14 – 15	16 – 17
No. of mutual funds	20	40	30	25	15

Solution:

The given classes are not continuous. Lets make the classes continuous.

Continuous Classes	7.5 – 9.5	9.5 – 11.5	11.5 – 13.5	13.5 – 15.5	15.5 – 17.5
Frequency	20	40	30	25	15



(ii) Frequency Polygon:

There are two methods to draw frequency polygon or curve:

(i) Using histogram

(ii) Plotting the points.

In the first method, we have to draw histogram and then mark all the mid-points on the upper side of the rectangles and then we join all the points with scale. To close the polygon or curve, we have join the midpoint of the class before the first class and the mid-point of the class after the last class.

In the second method, we plot the points on the graph paper with midpoint of class as x co-ordinate and frequency as y co-ordinate (midpoint, frequency) and then join all the points in order with scale and then close the polygon or curve as mentioned in the previous method.



➤ **Steps to draw frequency polygon or frequency curve:**

Step-1: Prepare frequency distribution table with class marks.

Step-2: Draw X-axis, Y-axis, choose proper scale.

Step-3: Take class marks on X-axis.

Step-4: Take frequencies (f_i) on Y-axis

Step-5: Plot the points (x_i, f_i).

Step-6: Plot two additional points with two additional classes one preceding the first class with frequency zero (0) and the other succeeding the last class with frequency zero (0).

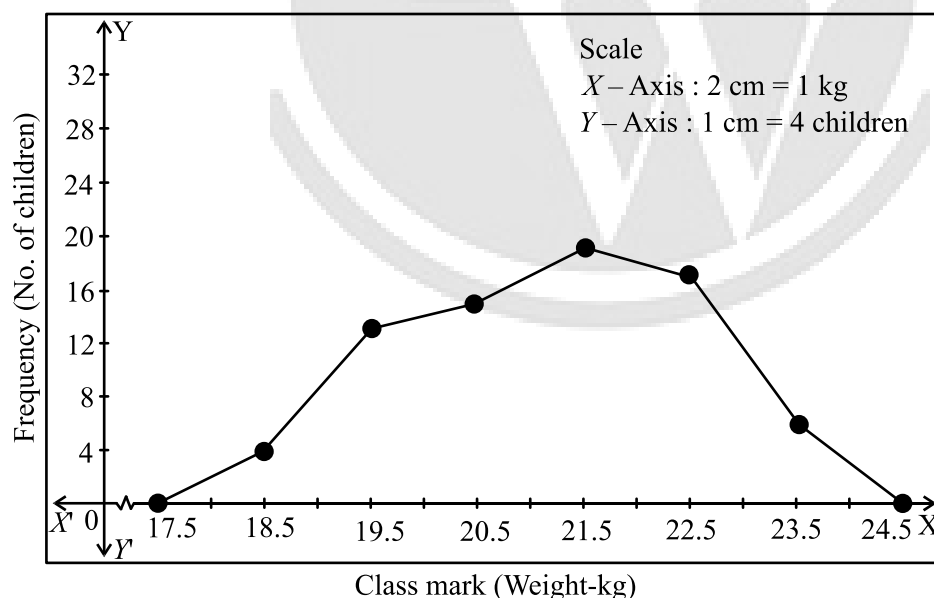
Example:

The following table shows the weights of children and the number of children. Draw a frequency polygon showing the information.

Weight of children (kg)	18 – 19	19 – 20	20 – 21	21 – 22	22 – 23	23 – 24
No. of children	4	13	15	19	17	6

Let us prepare a table showing the co-ordinates necessary to draw a frequency polygon.

Class	18 – 19	19 – 20	20 – 21	21 – 22	22 – 23	23 – 24
Class mark	18.5	19.5	20.5	21.5	22.5	23.5
Frequency	4	13	15	19	17	6
Coordinates of points	(18.5, 4)	(19.5, 13)	(20.5, 15)	(21.5, 19)	(22.5, 17)	(23.5, 6)



(iii) Pie diagram:

The relative values of items are represented by a sector of circle. Since the sectors resembles the slices of pie, therefore it is called a pie diagram.

The main terms related to pie diagrams are sector and central angle.

Steps to draw a pie diagram:

Step-1: Find the total of all values of items or that of components corresponds to 360° . Hence, find the measure of central angle (θ) for each given information.

Step-2: Draw a circle of convenient radius (usually within 4 – 5 cm range preferred.)



Step-3: Divide the circle into number of sectors using the value of central angle (θ) to represent the various items or components.

$$\text{Central angle } (\theta) = \frac{\text{value of the component}}{\text{Total value}} \times 360^\circ.$$

Activity:

The monthly expenditure of a family of different items is shown in the following table. Calculate the related central angle and the related central angles and draw a pie chart.

Pie Diagram:

Different Items	% of Expenditure	Measure of Central angle
Food	40	$\frac{40}{100} \times 360 = 144^\circ$
Clothing	20	$\frac{20}{100} \times \frac{360}{360} = 72^\circ$
House rent	15	$\frac{15}{100} \times \frac{360}{360} = 54^\circ$
Education	20	$\frac{20}{100} \times \frac{360}{360} = 72^\circ$
Expenditure	05	$\frac{05}{100} \times \frac{360}{360} = 18^\circ$
Total	100	360°

